WEEK#04 PROJECT REPORT

Outbreak Spread Predictor

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1. Introduction

The COVID-19 pandemic affected millions of people worldwide and generated a large amount of publicly available health data. Analyzing this data helps us understand infection trends and supports future planning.

This project focuses on analyzing the global COVID-19 confirmed cases dataset and predicting future outbreak trends using a Machine Learning model. The project also includes data visualization techniques to better understand the spread of the disease over time.

Python was used as the primary programming language because of its powerful libraries for data analysis, visualization, and machine learning.

2. Project Objectives

The main objectives of this project are:

- To analyze COVID-19 confirmed cases data.
- To clean and organize the dataset.
- To visualize outbreak trends using graphs.
- To identify patterns in the available data.
- To build a Machine Learning model for prediction.
- To create a simple graphical dashboard using PyQt5.

3. Tools and Technologies Used

Tool	***Purpose***
Python	Programming Language
Pandas	Data Analysis
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning
Joblib	Saving Machine Learning Model
PyQt5	Desktop Dashboard
Jupyter Notebook	Development Environment
GitHub	Project Repository

4. Dataset Description

The dataset used in this project contains worldwide COVID-19 confirmed case records.

Main dataset files include:

- time_series_covid19_confirmed_global.csv
- time_series_covid19_deaths_global.csv
- time_series_covid19_recovered_global.csv

The dataset contains:

- Country/Region
- Province/State
- Latitude
- Longitude
- Daily confirmed case counts

The data was loaded using the Pandas library and processed for analysis.

5. Project Methodology

The project was completed in the following stages.

Step 1: Data Collection

The COVID-19 dataset was downloaded and stored inside the project.

Step 2: Data Loading

The dataset was imported using Pandas.

Step 3: Data Exploration

The dataset was examined using:

- head()
- info()
- describe()
- shape
- columns

Step 4: Data Cleaning

Missing values were checked and the cleaned dataset was prepared for further analysis.

Step 5: Data Visualization

Different graphs were created to understand outbreak trends.

Step 6: Machine Learning

A Linear Regression model was trained using Scikit-learn to predict future confirmed cases.

Step 7: Dashboard

A simple desktop dashboard was created using PyQt5.

6. Data Visualization

Several visualizations were developed to understand the COVID-19 outbreak.

These include:

- Line Chart showing confirmed cases over time.
- Bar Chart for country-wise comparison.
- Heatmap displaying case intensity.
- Prediction Graph showing future outbreak trends.

7. Machine Learning Model

A Linear Regression algorithm was used to predict future COVID-19 confirmed cases.

The workflow included:

- Preparing the dataset
- Splitting the data
- Training the model
- Making predictions
- Evaluating prediction performance

The trained model was saved using the Joblib library for future use.

8. Dashboard

A desktop dashboard was developed using PyQt5.

The dashboard provides a simple graphical interface for displaying project information and can be extended to visualize prediction results in future versions.

9. Results

The project successfully achieved its objectives.

The dataset was analyzed successfully.

Visualizations clearly represented the outbreak trends.

The Machine Learning model successfully generated predictions based on historical data.

The project demonstrates how data analysis and machine learning can be combined to understand disease spread using publicly available datasets.

10. Conclusion

This project provided practical experience in the complete data science workflow, including data collection, preprocessing, visualization, model development, and deployment of a simple desktop interface.

Working on this project improved my understanding of Python libraries, machine learning concepts, and data visualization techniques.

The project also demonstrated the importance of data-driven decision making in analyzing public health information.